

● EASY

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Geometry and Trigonometry

01

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**A****A. 324**

Choice A is correct. It's given that triangle JKL is similar to triangle RST , where J corresponds to R and K corresponds to S . It follows that $\bar{JK}$ corresponds to $\bar{RS}$. If two triangles are similar, then the scale factor between their perimeters is equal to the scale factor between the lengths of their corresponding sides. It's given that the length of $\bar{JK}$ is 15 and the length of $\bar{RS}$ is 135. Therefore, the scale factor from the length of $\bar{JK}$ to the length of $\bar{RS}$ is $\frac{135}{15}$, or 9. It's given that the perimeter of triangle JKL is 36. Let p represent the perimeter of triangle RST . It follows that $\frac{p}{36} = 9$. Multiplying each side of this equation by 36 yields $p = 324$. Therefore, the perimeter of triangle RST is 324.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**C****C. 116**

Choice C is correct. Since lines ℓ and k are parallel, corresponding angles formed by the intersection of line j with lines ℓ and k are congruent. Therefore, the angle with measure a° must be the supplement of the angle with measure 64° . The sum of two supplementary angles is 180° , so $a = 180 - 64 = 116$.

Choice A is incorrect and likely results from thinking the angle with measure a° is the complement of the angle with measure 64° . Choice B is incorrect and likely results from thinking the angle with measure a° is congruent to the angle with measure 64° . Choice D is incorrect and likely results from a conceptual or computational error.

Q3**B****B. 10.8**

Choice B is correct. By the Pythagorean theorem, if a right triangle has a hypotenuse with length t and legs with lengths r and s , then $r^2 + s^2 = t^2$. It's given in the right triangle shown that the legs have lengths of 9 and 6 and the hypotenuse has a length of c . Substituting 9 for r , 6 for s , and c for t in $r^2 + s^2 = t^2$ yields $9^2 + 6^2 = c^2$, or $117 = c^2$. Taking the square root of both sides of this equation yields $\pm\sqrt{117} = c$. Since the length of a side of a triangle must be positive, the value of c is $\sqrt{117}$, which is approximately equal to 10.8167. Of the choices, 10.8 is the closest to the value of c .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**A**

A. 9

Choice A is correct. A circle has 360 degrees of arc. In the circle shown, O is the center of the circle and $\angle AOC$ is a central angle of the circle. From the figure, the two diameters that meet to form $\angle AOC$ are perpendicular, so the measure of $\angle AOC$ is 90° . Therefore, the length of minor arc $\widehat{AC}$ is $\frac{90}{360}$ of the circumference of the circle. Since the circumference of the circle is 36, the length of minor arc $\widehat{AC}$ is $\frac{90}{360} \times 36 = 9$.

Choices B, C, and D are incorrect. The perpendicular diameters divide the circumference of the circle into four equal arcs; therefore, minor arc $\widehat{AC}$ is $\frac{1}{4}$ of the circumference. However, the lengths in choices B and C are, respectively, $\frac{1}{3}$ and $\frac{1}{2}$ the circumference of the circle, and the length in choice D is the length of the entire circumference. None of these lengths is $\frac{1}{4}$ the circumference.

Q5**180**

The correct answer is 180. The perimeter of a polygon is equal to the sum of the lengths of the sides of the polygon. It's given that each side of the square has a length of 45. Since a square is a polygon with 4 sides, the perimeter of this square is $45 + 45 + 45 + 45$, or 180.

Q6**B****B.** 39

Choice B is correct. The sum of the measures of the interior angles of a triangle is 180 degrees. Since the triangle is a right triangle, it has one angle that measures 90 degrees. Therefore, the sum of the measures, in degrees, of the remaining two angles is $180 - 90$, or 90. It's given that the measure of one of the acute angles in the triangle is 51 degrees. Therefore, the measure, in degrees, of the other acute angle is $90 - 51$, or 39.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the measure, in degrees, of the acute angle whose measure is given.

Q7**B****B.** $\sqrt{202}$

Choice B is correct. The Pythagorean theorem states that for a right triangle, $c^2 = a^2 + b^2$, where c represents the length of the hypotenuse and a and b represent the lengths of the legs. It's given that a right triangle has legs with lengths of 11 centimeters and 9 centimeters.

Substituting 11 for a and 9 for b in the formula $c^2 = a^2 + b^2$ yields $c^2 = 11^2 + 9^2$, which is equivalent to $c^2 = 121 + 81$, or $c^2 = 202$. Taking the square root of each side of this equation yields $c = \pm \sqrt{202}$. Since c represents a length, c must be positive. Therefore, the length of the triangle's hypotenuse, in centimeters, is $\sqrt{202}$.

Choice A is incorrect. This is the result of solving the equation $c^2 = 112 + 92$, not $c^2 = 11^2 + 9^2$.

Choice C is incorrect. This is the result of solving the equation $c^2 = 112 + 92$, not $c^2 = 11^2 + 9^2$.

Choice D is incorrect. This is the result of solving the equation $c = 11^2 + 9^2$, not $c^2 = 11^2 + 9^2$.

Q8**C**

C. $\frac{\sqrt{3}}{2}$

Choice C is correct. The cosine of an angle is equal to the cosine of $n(2\pi)$ radians more than the angle, where n is an integer constant. Since $\frac{565\pi}{6}$ is equivalent to $47\left(2\pi\right) + \frac{\pi}{6}$, $\cos\left(\frac{565\pi}{6}\right)$ can be rewritten as $\cos\left(47\left(2\pi\right) + \frac{\pi}{6}\right)$, which is equal to $\cos\left(\frac{\pi}{6}\right)$. Therefore, the value of $\cos\left(\frac{565\pi}{6}\right)$ is equal to the value of $\cos\left(\frac{\pi}{6}\right)$, which is $\frac{\sqrt{3}}{2}$.

Alternate approach: A trigonometric ratio can be found using the unit circle, that is, a circle with radius 1 unit. The cosine of a number t is the x -coordinate of the point resulting from traveling a distance of t counterclockwise from the point $(1, 0)$ around a unit circle centered at the origin in the xy -plane. A unit circle has a circumference of 2π . It follows that since $\frac{565\pi}{6}$ is equal to $47\left(2\pi\right) + \frac{\pi}{6}$, traveling a distance of $\frac{565\pi}{6}$ counterclockwise around a unit circle means traveling around the circle completely 47 times and then another $\frac{\pi}{6}$ beyond that. That is, traveling $\frac{565\pi}{6}$ results in the same point as traveling $\frac{\pi}{6}$. Traveling $\frac{\pi}{6}$ counterclockwise from the point $(1, 0)$ around a unit circle centered at the origin in the xy -plane results in the point $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$. Thus, the value of $\cos\frac{565\pi}{6}$ is the x -coordinate of the point $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$, which is $\frac{\sqrt{3}}{2}$.

Choice A is incorrect. This is the value of $\sin\frac{565\pi}{6}$, not $\cos\frac{565\pi}{6}$.

Choice B is incorrect. This is the value of the cosine of a multiple of 2π , not $\frac{565\pi}{6}$.

Choice D is incorrect. This is the value of $\frac{1}{\tan\frac{565\pi}{6}}$, not $\cos\frac{565\pi}{6}$.

Q9**1800**

The correct answer is 1,800. The area, A , of a triangle can be found using the formula $A = \frac{1}{2}bh$, where b is the base length of the triangle and h is the height of the triangle. It's given that the triangle has a base length of 40 centimeters and a height of 90 centimeters. Substituting 40 for b and 90 for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}4090$, or $A = 1,800$. Therefore, the area, in square centimeters, of the triangle is 1,800.

Q10

B

B. 120

Choice B is correct. The sum of the interior angles of a triangle is 180° . The measures of the two interior angles of the given triangle are shown. Therefore, the measure of the third interior angle is $180^\circ - 70^\circ - 50^\circ = 60^\circ$. The angles of measures x° and 60° are supplementary, so their sum is 180° . Therefore, $x = 180 - 60 = 120$.

Choice A is incorrect and may be the result of misinterpreting x° as supplementary to 70° .

Choice C is incorrect and may be the result of misinterpreting x° as supplementary to 50° .

Choice D is incorrect and may be the result of a calculation error.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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